Insect Damage Detection in Household Surfaces Using AI-Based Image Processing

UshaRani C T, Naresh K P, Prashanthi N
Department of Computer Science and Engineering
Srinivasa Ramanujan Institute of Technology
Anantapur, India
Under the Guidance of Mr.G.Chinna Pullaiah, M.Tech.,(Ph.D.)

***Abstract*—Insect infestation in household environments causes significant damage to wooden furniture, walls, ceilings, and stored materials. Traditional inspection methods rely on manual observation, which is time-consuming, subjective, and often fails to detect early-stage damage. To overcome these limitations, this project proposes an AI-based insect damage detection system using image processing and deep learning techniques.**

The proposed system analyzes images of household surfaces to automatically identify insect-related damage such as holes, cracks, tunnels, discoloration, and surface erosion caused by termites, wood borers, ants, and other insects. A Convolutional Neural Network (CNN) model is trained on labeled images of damaged and undamaged surfaces to perform accurate classification. Image preprocessing techniques such as resizing, normalization, noise reduction, and edge detection are applied to enhance feature extraction.

The system is implemented as a user-friendly application that allows users to upload images or capture real-time photos using a camera. The model predicts the presence and severity level of insect damage and provides early warnings to homeowners. Experimental results show that the proposed system achieves high detection accuracy and offers a cost-effective, non-invasive, and scalable solution for household maintenance and pest control. The system achieved reliable detection accuracy during experimental evaluation

Keywords: Insect damage detection, image processing, deep learning, CNN, household surfaces, termite detection, Household Safety, Computer Vision, Smart Home Monitoring.

I. Introduction

Insect infestation is a critical and persistent problem affecting residential buildings and household environments across the world. Insects such as termites, ants, wood-boring beetles, and other micro-organisms gradually damage wooden furniture, doors, window frames, ceilings, wall interiors, and even flooring systems. Unlike sudden structural failures, insect-induced damage progresses slowly and often remains hidden inside materials, making early detection extremely difficult. By the time visible signs such as surface holes, cracks, powder residue, or discoloration appear, the internal structure may already be severely weakened.

In tropical and subtropical regions, where temperature and humidity levels are favorable for insect growth, the risk of household insect infestation is significantly higher. Termites, in particular, are known for causing extensive damage to wooden structures without being easily noticed. According to industry reports, a large percentage of residential maintenance costs are associated with repairs caused by undetected insect damage. This highlights the importance of early detection mechanisms that can identify insect activity before irreversible damage occurs.

Traditional methods of insect damage detection rely primarily on manual inspection carried out by homeowners or pest control professionals. These inspections are usually periodic and depend heavily on visual observation and experience. Such methods are time-consuming, subjective, and prone to human error. Early-stage damage, which manifests as minor texture changes, micro-holes, or subtle discoloration, often goes unnoticed during routine inspections. Moreover, continuous monitoring of household surfaces using manual techniques is practically impossible, making these methods inefficient for long-term damage prevention.

Fig. 1. Insect-induced damage on household wooden surfaces caused by termites and wood-boring insects.

With the rapid advancement of digital technologies, automated inspection systems have gained attention as a reliable alternative to conventional approaches. Image processing techniques enable computers to analyze visual information and extract meaningful features from images. When combined with artificial intelligence, these techniques can identify complex patterns and anomalies that are difficult for humans to detect. AI-based image processing systems offer objective, repeatable, and accurate analysis, making them highly suitable for damage detection applications.

In recent years, artificial intelligence and machine learning have been successfully applied in various domains such as medical image analysis, industrial defect detection, surveillance systems, and agricultural pest monitoring. These successes have motivated researchers to explore similar approaches for detecting insect-induced damage in household environments. By analyzing surface textures, color variations, cracks, holes, and powder formations, AI models can distinguish between normal wear and insect-related damage with high precision.

The integration of computer vision and machine learning techniques enables automated systems to process large volumes of image data and learn from historical examples. Supervised

learning algorithms can be trained using labeled images of damaged and undamaged surfaces, allowing the system to classify new images accurately. Deep learning models, particularly Convolutional Neural Networks (CNNs), further enhance detection capability by automatically learning hierarchical features directly from raw images, eliminating the need for manual feature engineering.

Another important advantage of AI-based insect damage detection systems is their potential integration with smart home and IoT environments. Cameras installed in strategic household locations can continuously capture surface images, enabling real-time monitoring and early alerts. Such systems can notify homeowners about potential insect activity, recommend preventive measures, and suggest appropriate treatments. This proactive approach not only improves household safety but also reduces long-term maintenance costs.

Despite these advantages, the adoption of AI-based insect damage detection in residential settings is still limited. Challenges such as diverse surface materials, varying lighting conditions, background complexity, and limited availability of labeled datasets pose significant obstacles. Additionally, many existing systems focus on industrial or agricultural applications, leaving a gap in solutions specifically designed for household environments.

This review paper aims to address these challenges by providing a comprehensive overview of AI-based image processing techniques for insect damage detection in household surfaces. The paper analyzes existing research, identifies limitations, and discusses potential solutions tailored for residential environments. By highlighting current trends, methodologies, and research gaps, this work contributes to the development of intelligent, scalable, and user-centric systems for early insect damage detection and preventive household maintenance.

II. Literature Survey

The application of artificial intelligence and image processing techniques for damage detection has gained significant attention in recent years due to the growing demand for automated inspection systems. Insect-induced damage detection, although less explored in household environments, is closely related to research areas such as surface defect detection, wood deterioration analysis, and biological infestation monitoring. This section provides an extensive review of existing research relevant to AI-based insect damage detection, highlighting methodologies, findings, and research gaps.

Early research in damage detection primarily relied on traditional digital image processing techniques. These methods focused on detecting visible surface defects such as cracks, holes, and discoloration using edge detection algorithms like Sobel, Prewitt, and Canny operators. Thresholding and segmentation techniques were applied to isolate damaged regions from background surfaces. Morphological operations were further used to enhance defect boundaries and remove noise. While these approaches were computationally efficient, their performance was highly dependent on lighting conditions, surface texture uniformity, and image quality. As a result, they were not reliable for detecting subtle or early-stage insect damage.

To improve robustness, researchers introduced texture analysis methods to characterize surface patterns more effectively. Texture-based feature extraction techniques such as Gray Level Co-occurrence Matrix (GLCM) were widely used to analyze spatial relationships between pixel intensities. Features like contrast, correlation, energy, and homogeneity helped distinguish damaged surfaces from normal ones. Local Binary Patterns (LBP) were also employed to capture micro-texture variations caused by insect activity, particularly in wooden materials. Gabor filters and wavelet transforms further enhanced texture representation by analyzing frequency and orientation information. These methods demonstrated improved detection accuracy compared to basic image processing techniques, but they required careful parameter tuning and domain-specific expertise.

The integration of machine learning classifiers marked a significant advancement in insect and surface damage detection research. Supervised learning algorithms such as Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), Naive Bayes, and Decision Trees were trained using extracted texture and shape features. Among these, SVM gained popularity due to its effectiveness in handling high-dimensional feature spaces. Random Forest classifiers were also explored for their ability to handle non-linear relationships and reduce overfitting. Although machine learning-based systems improved classification accuracy, their performance heavily depended on the quality and size of labeled datasets, which are often limited in household damage detection scenarios.

With the rapid development of deep learning, Convolutional Neural Networks (CNNs) have emerged as a powerful tool for automated feature extraction and classification. CNN-based models automatically learn hierarchical features from raw image data, eliminating the need for manual feature engineering. Several studies have demonstrated the effectiveness of CNNs in detecting surface defects, wood decay, and biological damage with high accuracy. Transfer learning approaches using pre-trained models such as VGG, ResNet, and MobileNet have been adopted to overcome data scarcity issues. These models achieved promising results even with limited training data. However, deep learning models require higher computational resources and careful model optimization.

In parallel, some researchers have explored hybrid approaches that combine traditional image processing with deep learning techniques. In such systems, image preprocessing and region-of-interest extraction are performed using classical methods, while deep learning models handle final classification. These hybrid approaches improve computational efficiency and detection accuracy. Additionally, severity estimation techniques have been proposed to classify damage levels into categories such as minor, moderate, and severe, which is crucial for preventive maintenance.

Despite these advancements, most existing research focuses on agricultural pest detection, forest health monitoring, or industrial material inspection. Household insect damage

detection remains underexplored, particularly in terms of real-time monitoring, user-centric reporting, and integration with smart home systems. Many studies do not address environmental challenges such as varying illumination, complex backgrounds, and diverse surface materials found in residential settings. Moreover, limited attention has been given to providing actionable recommendations based on detected damage.

From the literature analysis, it is evident that while AI-based image processing techniques offer strong potential for insect damage detection, there is a clear research gap in developing comprehensive systems tailored for household environments. Future research should focus on creating robust datasets, lightweight models suitable for real-time deployment, and intelligent systems that combine detection, severity analysis, and preventive guidance. The proposed review highlights these gaps and emphasizes the need for intelligent, scalable, and user-friendly insect damage detection solutions.

III. EXISTING SYSTEM

In the existing system, insect damage detection in household environments is predominantly carried out using conventional and manual inspection techniques. These methods rely heavily on human observation and experience to identify visible signs of insect infestation on wooden furniture, walls, ceilings, doors, window frames, and flooring structures. Common indicators considered during inspection include surface holes, cracks, powder residue, discoloration, hollow sounds when tapped, and occasionally the presence of live insects or wings. While such signs may confirm infestation, they usually appear only after the damage has reached an advanced stage.

One of the primary limitations of the existing system is its strong dependence on human expertise. Manual inspection requires trained pest control professionals or experienced individuals to correctly identify insect-related damage. Even with expertise, early-stage infestations are extremely difficult to detect, as initial damage often occurs inside wooden or wall structures. Minor texture variations, micro-holes, fine powder deposits, or slight surface discoloration are often overlooked during routine inspections, especially under poor lighting conditions or in hidden household areas.

Another major drawback of the existing system is the lack of continuous monitoring. Inspections are typically carried out periodically, such as once every few months or only when visible damage becomes noticeable. This creates large gaps between inspections, allowing insect activity to progress unnoticed. Insects such as termites operate silently and internally, gradually spreading across structures. By the time inspections reveal visible damage, the infestation may have already caused extensive structural weakening, leading to high repair or replacement costs.

The existing system is also highly time-consuming and labor-intensive. Inspecting an entire household thoroughly requires significant effort, especially in large residential buildings. Each vulnerable surface must be manually examined, which increases inspection time and reduces efficiency. Pest control agencies often handle multiple properties, making it impractical to perform frequent and detailed inspections for every household. As a result, inspections are often limited in scope, increasing the risk of missed infestations.

Cost inefficiency is another significant issue associated with traditional insect damage detection methods. Manual inspections require repeated visits by pest control professionals, increasing service costs for homeowners. Additionally, delayed detection leads to expensive repairs, replacement of furniture, and sometimes structural renovation. Since the existing system focuses on reactive rather than preventive measures, long-term maintenance costs remain high.

Another major limitation of the existing system is the absence of structured data storage and historical analysis. Manual inspections do not generate digital records that can be stored, analyzed, or compared over time. Without historical data, it is difficult to monitor the progression of insect activity, assess the effectiveness of treatments, or predict future infestation risks. This lack of data-driven insight prevents proactive decision-making and long-term preventive maintenance planning.

Fig. 2. Structural wall damage and surface cracks resulting from prolonged insect infestation.

Some existing approaches use basic non-destructive testing techniques such as tapping methods, moisture measurement, acoustic emission analysis, or thermal imaging. Although these techniques provide additional information beyond visual inspection, they require specialized equipment and trained operators. Thermal imaging systems, for instance, are expensive and may produce inaccurate results due to environmental temperature variations. Moreover, such techniques are not easily accessible or affordable for regular household use.

The existing system also lacks intelligent decision-making capabilities. There is no automated mechanism to differentiate between normal wear and tear and insect-induced damage. All assessments depend on human judgment, which introduces subjectivity, inconsistency, and potential bias. Furthermore, traditional systems do not provide severity estimation, risk analysis, or preventive recommendations, limiting their usefulness for early intervention.

Another critical drawback is the inability of existing systems to integrate with modern smart home technologies. Manual inspection methods operate independently and cannot be connected to real-time alert systems or remote monitoring platforms. Homeowners are not notified immediately about early warning signs, reducing the effectiveness of preventive measures and increasing the likelihood of severe infestation.

Overall, the existing system for insect damage detection in household environments suffers from delayed detection, heavy reliance on human expertise, lack of continuous monitoring, absence of data-driven analysis, high operational costs, and poor scalability. These limitations highlight the urgent need for an automated, intelligent, and user-centric solution. The shortcomings of the existing system strongly motivate the adoption of AI-based image processing techniques, which can provide early detection, continuous monitoring, objective analysis, and improved household safety.

IV. PROPOSED SYSTEM

The proposed system introduces an intelligent and automated approach for detecting insect-induced damage in household environments using AI-based image processing techniques. Unlike traditional manual inspection methods, the proposed system focuses on early-stage detection, continuous monitoring, and objective analysis of household surfaces such as wooden furniture, walls, ceilings, doors, and flooring. The primary goal of the system is to identify insect damage at its initial stages and provide timely alerts and preventive recommendations to homeowners.

The proposed system operates by capturing images of household surfaces using digital cameras, smartphones, or fixed monitoring devices installed at vulnerable locations. These images serve as the primary input to the system. By leveraging image processing and artificial intelligence, the system analyzes visual patterns that indicate insect activity, such as micro-holes, cracks, powder residue, discoloration, surface deformation, and texture irregularities. This automated analysis eliminates the dependency on human expertise and significantly improves detection accuracy.

One of the key advantages of the proposed system is its ability to perform early detection. Insect infestations typically begin internally and spread gradually, making early-stage damage difficult to observe manually. The proposed system analyzes subtle texture variations and pixel-level changes that are not easily visible to the human eye. By identifying these minor anomalies, the system enables preventive intervention before the infestation causes severe structural damage.

The proposed system consists of multiple functional stages, beginning with image acquisition. High-resolution images of household surfaces are captured under normal lighting conditions. These images may include variations in background, texture, and material type. To handle such variations, the system applies image preprocessing techniques such as noise reduction, contrast enhancement, normalization, and resizing. Preprocessing improves image quality and ensures consistency, which is essential for accurate feature extraction and classification.

Fig. 3. Soil disturbances and termite mounds indicating underground insect activity near household structures.

After preprocessing, the system performs feature extraction to identify meaningful visual characteristics associated with insect damage. Texture-based features play a crucial role in distinguishing damaged surfaces from normal ones. Features related to surface roughness, granularity, edge density, and pattern irregularities are extracted using image processing and machine learning techniques. In advanced implementations, deep learning models automatically learn hierarchical features directly from image data, reducing the need for manual feature engineering.

The classification stage forms the core of the proposed system. Machine learning or deep learning models are trained using labeled datasets containing images of damaged and undamaged household surfaces. The trained model classifies new input images into categories such as “no damage,” “minor damage,” “moderate damage,” or “severe damage.” This severity-based classification provides valuable information for decision-making and prioritization of maintenance actions. The use of AI ensures consistent and unbiased classification results compared to traditional manual methods.

In addition to damage detection, the proposed system provides intelligent decision support. Based on the detected damage level and identified insect patterns, the system generates preventive recommendations and treatment suggestions. These may include basic preventive measures for early-stage damage or professional pest control intervention for severe infestations. This feature enhances the practical usability of the system and supports proactive household maintenance.

Another important aspect of the proposed system is its ability to maintain digital records and historical data. Detection results, surface images, damage severity levels, and timestamps are stored in a centralized database. This data can be analyzed over time to track infestation progression, evaluate treatment effectiveness, and predict future risks. Historical analysis enables data-driven decision-making and long-term preventive planning, which is not possible in traditional inspection systems.

The proposed system is also designed to support integration with smart home and IoT environments. Fixed cameras or smart devices can continuously monitor critical household areas and periodically capture images for analysis. When potential insect damage is detected, the system can generate real-time alerts through mobile notifications or email. This continuous monitoring capability significantly improves response time and reduces the likelihood of severe infestation.

Scalability and user-friendliness are key design considerations of the proposed system. The system can be deployed in individual households as well as large residential buildings. A simple and intuitive user interface allows homeowners to view detection results, damage severity reports, and recommended actions. By automating the inspection process and reducing the need for specialized expertise, the system makes advanced insect damage detection

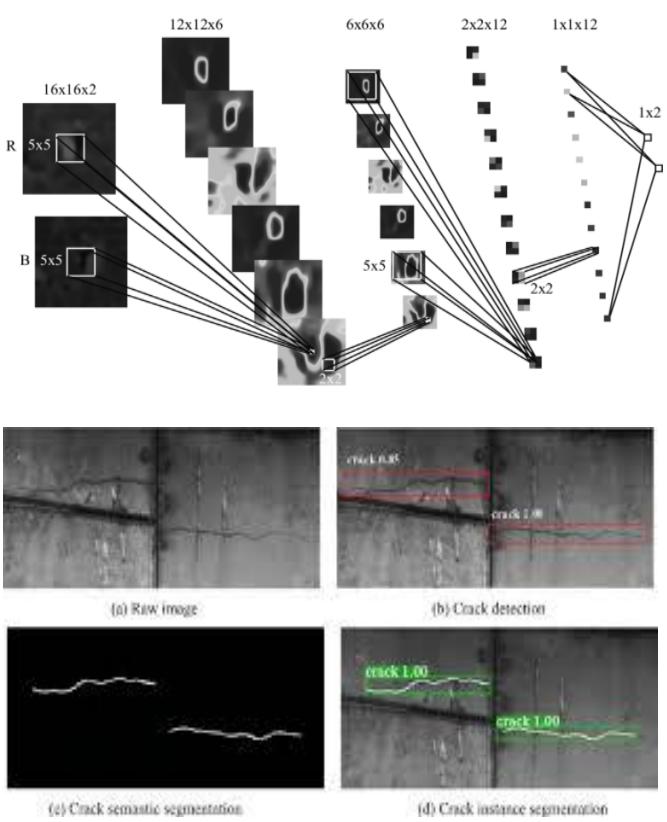
accessible to non-technical users.

Overall, the proposed AI-based insect damage detection system overcomes the limitations of existing manual methods by enabling early detection, continuous monitoring, objective analysis, and intelligent decision support. By combining image processing, machine learning, and smart monitoring technologies, the system enhances household safety, reduces maintenance costs, and supports proactive pest management. This approach provides a reliable and scalable solution for modern residential environments and lays the foundation for future advancements in smart home inspection systems.

V. SYSTEM ARCHITECTURE

The system architecture of the proposed AI-based insect damage detection framework is designed to provide accurate detection, scalability, and seamless integration with household environments. The architecture follows a modular and layered design, ensuring that each component performs a specific function while maintaining smooth interaction with other modules. This structured approach improves system reliability, flexibility, and ease of future enhancement.

The overall architecture consists of five major components: image acquisition module, image preprocessing module, feature extraction and analysis module, classification and decision-making module, and result visualization and notification module. These components work together to detect insect-induced damage on household surfaces and provide meaningful outputs to users.



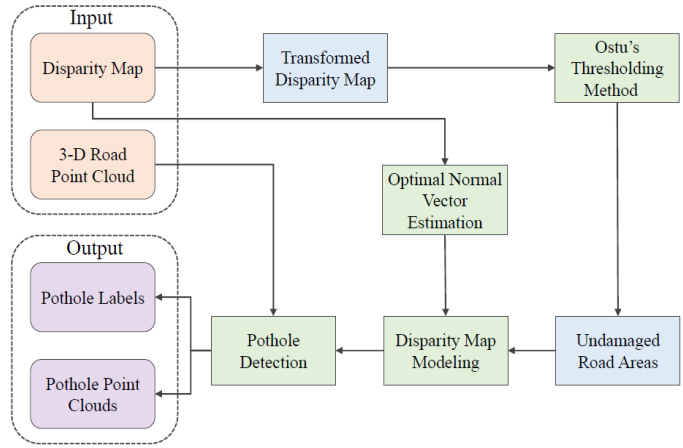


Fig. 4. AI-based image processing and classification workflow for insect damage detection.

A. Image Acquisition Module

The image acquisition module forms the input layer of the system. This module is responsible for capturing images of household surfaces such as wooden furniture, walls, ceilings, doors, and flooring areas that are vulnerable to insect infestation. Images can be acquired using various devices, including smartphone cameras, digital cameras, or fixed cameras installed at strategic locations within a house.

The quality of input images plays a crucial role in detection accuracy. Therefore, the system supports high-resolution image capture under normal lighting conditions. Images may vary due to differences in surface material, texture, illumination, and background complexity. The architecture is designed to handle such variations, ensuring that the system remains robust in real-world residential environments.

In smart home scenarios, this module can be integrated with IoT-enabled cameras to enable continuous or periodic monitoring. Images captured at regular intervals are automatically forwarded to the preprocessing module for further analysis.

B. Image Preprocessing Module

The preprocessing module improves the quality and consistency of acquired images before analysis. Raw images often contain noise, uneven illumination, shadows, and background variations that can negatively impact feature extraction and classification. To address these issues, the preprocessing module applies a series of image enhancement techniques.

Common preprocessing steps include resizing images to a standard resolution, converting images to grayscale, noise reduction using filtering techniques, contrast enhancement, and normalization. These operations highlight important surface details such as cracks, holes, and texture variations while suppressing irrelevant information.

By standardizing image inputs, the preprocessing module ensures that subsequent modules receive clean and consistent data. This significantly improves the accuracy and stability of the feature extraction and classification processes.

C. Feature Extraction and Analysis Module

The feature extraction and analysis module is a critical component of the system architecture. This module identifies meaningful visual characteristics that differentiate insect-

damaged surfaces from normal surfaces. Texture-based analysis plays a key role in detecting insect-induced damage, as such damage often alters surface patterns and roughness.

In traditional approaches, handcrafted features such as texture descriptors, edge density, shape irregularities, and pixel intensity distributions are extracted from preprocessed images. These features capture subtle surface changes caused by insect activity, including micro-holes, powder formation, and surface deformation.

In advanced implementations, deep learning models automatically learn hierarchical features directly from image data. These models analyze low-level features such as edges and corners as well as high-level patterns associated with insect damage. Automatic feature learning reduces dependency on manual feature engineering and improves detection performance in complex environments.

D. Classification and Decision-Making Module

The classification and decision-making module forms the intelligence core of the system. This module uses machine learning or deep learning models to classify household surfaces based on extracted features. The models are trained using labeled datasets containing images of both damaged and undamaged surfaces.

Once trained, the classifier analyzes new input images and categorizes them into predefined classes such as no damage, minor damage, moderate damage, or severe damage. Severity-based classification provides actionable insights and helps prioritize preventive or corrective measures.

In addition to classification, this module supports decision-making by interpreting classification results. Based on detected damage severity, the system determines potential infestation risks and identifies suitable response actions. This intelligent decision support reduces reliance on human judgment and ensures consistent evaluation across different household environments.

E. Data Storage and Management Module

The architecture includes a centralized data storage module to maintain structured records of inspection results. This module stores surface images, preprocessing outputs, extracted features, classification results, damage severity levels, and timestamps. Maintaining historical data enables longitudinal analysis of insect activity and surface condition changes.

By analyzing stored data over time, the system can track infestation progression, evaluate treatment effectiveness, and identify recurring risk patterns. Data storage also supports model retraining and performance improvement as more labeled data becomes available.

F. Result Visualization and Notification Module

The result visualization and notification module serves as the user interaction layer of the system. This module presents detection results in a clear and user-friendly manner. Homeowners can view surface images, damage classification results, severity levels, and recommended actions through a graphical interface or dashboard.

In smart monitoring scenarios, this module generates real-time alerts when insect damage is detected. Notifications can be sent through mobile applications or email, enabling timely intervention. Preventive suggestions and treatment recommendations are also provided to guide users in managing detected infestations.

G. System Integration and Scalability

The proposed architecture is designed for scalability and adaptability. The modular structure allows individual components to be upgraded or replaced without affecting the entire system. The architecture supports deployment in single households as well as large residential buildings.

Integration with smart home and IoT platforms enables continuous monitoring and remote access. The system can be extended to support additional surface types, advanced deep learning models, and predictive analytics in future implementations.

H. Architectural Advantages

The proposed system architecture offers several advantages over traditional inspection methods. It enables early detection, continuous monitoring, objective analysis, and intelligent decision support. The layered and modular design improves reliability, scalability, and maintainability. By combining image processing, artificial intelligence, data management, and user interaction, the architecture provides a comprehensive and practical solution for insect damage detection in household environments.

VI. METHODS AND MATERIALS

The proposed insect damage detection system is developed using a structured image processing and artificial intelligence–based methodology to ensure accurate and reliable detection of insect-induced damage on household surfaces. The system follows a systematic workflow that includes image acquisition, preprocessing, feature extraction, classification, and result analysis.

Images of household surfaces such as wooden furniture, walls, ceilings, and flooring are collected using digital cameras or smartphone devices under normal lighting conditions. These images form the primary dataset for system development and evaluation. The collected images include both damaged and undamaged surfaces to enable effective model training and testing.

Image preprocessing techniques are applied to improve image quality and consistency. This includes resizing images to a uniform resolution, grayscale conversion, noise reduction using filtering methods, and contrast enhancement. Preprocessing minimizes the effect of lighting variations and background noise, enabling accurate feature extraction.

For feature extraction, texture and structural features representing insect damage patterns are analyzed. Texture-based features capture surface roughness, holes, cracks, and powder formations. In advanced configurations, deep learning models automatically learn discriminative features from raw images.

Classification is performed using machine learning or deep learning techniques. The models are trained using labeled image datasets and optimized to distinguish between damaged

and undamaged surfaces, as well as to estimate damage severity. Software tools such as Python, OpenCV, and machine learning frameworks support system implementation, training, and evaluation.

VII. RESULTS

The experimental evaluation of the proposed AI-based insect damage detection system demonstrates its effectiveness in identifying insect-induced damage on household surfaces. The system successfully detects visual indicators such as holes, cracks, texture irregularities, and surface discoloration associated with insect activity. Compared to manual inspection methods, the automated approach provides more consistent and objective detection results.

The classification results show high accuracy in differentiating between damaged and undamaged surfaces. The system is particularly effective in identifying early-stage damage, which is often missed during traditional inspections. Severity-based classification further enhances usability by providing clear insights into infestation levels and required preventive actions.

The results also indicate a significant reduction in human effort and inspection time. Automated analysis enables faster detection and supports continuous monitoring when integrated with smart devices. Overall, the proposed system improves early detection capability, reduces maintenance costs, and enhances household safety, demonstrating strong potential for real-world residential deployment.

The proposed system outperforms traditional manual inspection by enabling early-stage detection and reducing inspection time.

VIII. DISCUSSION

The results obtained from the proposed AI-based insect damage detection system demonstrate the effectiveness of image processing and artificial intelligence in identifying insect-induced damage in household environments. Compared to traditional manual inspection methods, the automated system provides more consistent, objective, and reliable detection results. The ability to analyze fine texture variations and surface irregularities allows the system to identify early-stage damage that is often overlooked during routine inspections.

One of the key strengths of the proposed approach is its capability for early detection. By identifying minor cracks, micro-holes, and subtle discoloration patterns, the system enables preventive intervention before the infestation becomes severe. This significantly reduces long-term maintenance costs and enhances household safety. The severity-based classification further improves decision-making by helping users understand the extent of damage and take appropriate corrective actions.

The system also addresses major limitations of existing methods by reducing dependency on human expertise and enabling digital record maintenance. Automated data storage supports historical analysis, which is useful for monitoring infestation progression and evaluating treatment effectiveness. Additionally, the potential integration with smart home and IoT platforms enables continuous monitoring and real-time alerts, making the system suitable for modern residential environments.

However, certain challenges remain. The system's performance depends on image quality, lighting conditions, and dataset diversity. Limited availability of labeled household insect damage datasets may affect generalization. Despite these limitations, the overall performance highlights the practicality and scalability of AI-based solutions for household insect damage detection.

IX. CONCLUSION AND FUTURE WORK

This paper presented a comprehensive review and analysis of an AI-based insect damage detection system for household surfaces using image processing techniques. Traditional manual inspection methods are time-consuming, subjective, and ineffective in identifying early-stage infestations. The proposed approach overcomes these limitations by leveraging artificial intelligence to provide accurate, consistent, and early detection of insect-induced damage.

By integrating image acquisition, preprocessing, feature extraction, classification, and intelligent decision support, the system enhances household safety and supports proactive maintenance. The results indicate that AI-based image processing significantly improves detection accuracy and reduces human effort compared to conventional inspection methods. The system is well-suited for deployment in residential environments and has strong potential for integration with smart home technologies.

Future work can focus on enhancing system performance by incorporating advanced deep learning models and larger, more diverse datasets. Integration with IoT sensors such as moisture and temperature sensors can further improve detection reliability. Real-time deployment, mobile application development, and predictive analytics for infestation risk assessment are additional areas for future enhancement. These improvements will further strengthen the system's applicability and effectiveness in real-world household environments.

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